

MultiSticker: Intent-Guided Multimodal Sticker Retrieval with Dialogue Memory

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Abstract

*Stickers are widely used in messaging platforms to express affect, humor and intent. Retrieving the right sticker for a dialogue turn is challenging because (i) the candidate bank is large and dominated by animated formats, (ii) the correct reply depends on long-range conversational state rather than just the last turn, and (iii) each context typically has only one observed positive even though many semantically equivalent stickers exist. We present **MultiSticker**, an intent-guided sticker retrieval system that combines three text streams—recent dialogue context, retrieved long-term session memory, and an LLM-generated reply intent—against a frozen-or-adapted OpenCLIP image bank covering .png, .gif and .webm stickers. We compare a frozen-CLIP zero-shot baseline against a learned retrieval head and three parameter-efficient fine-tuning variants (text-side LoRA, image-side LoRA, dual LoRA) on the U-Sticker corpus (8,129 sessions, 42,319 stickers). Text-side LoRA gives the largest improvement on exact top-1 retrieval, while image-side LoRA most improves semantic-group retrieval, indicating that the text and visual domains contribute complementary forms of adaptation. We argue that for a candidate-panel use case, semantic-group Hit@30 is the most meaningful target.*

1. Introduction

When users reply with stickers in a chat, the choice depends on local context (what was just said), long-range state (who is talking, prior tone, recurring topics) and the communicative *intent* behind the reply (e.g. gratitude, encouragement, teasing). A retrieval system that only looks at the last few turns frequently selects stickers with the wrong affect or the wrong recurring style for a given user stream. Conversely, a system that uses only image-side similarity ignores the dialogue entirely.

We focus on the U-Sticker corpus, which contains real multilingual chats spanning crypto, gaming and social-

domain channels. Several properties make sticker retrieval different from generic image-text retrieval. First, the candidate set is sparse and animated: 42,319 decodable stickers across .png, .gif and .webm. Second, supervision is *single-positive*: each turn records the one sticker the user actually sent, even though many other stickers in the bank are equally valid. Third, turns frequently contain code-switching and chat slang outside CLIP’s pre-training distribution.

We build on the IGSR-style design [7] of using a CLIP backbone plus a small learned head, and extend it in three directions:

- A retrieved *long-term memory* stream from previous sessions of the same user/file, encoded by multilingual E5 [8].
- An LLM-generated *reply intent* stream that abstracts away from the exact sticker id and describes the communicative function of the reply.
- A unified pipeline that handles .png, .gif and .webm via per-frame OpenCLIP encoding and mean pooling, and supports parameter-efficient adaptation through LoRA [2] on the text side, image side, or both.

Contributions. (1) A clean five-row experiment matrix (zero-shot CLIP, learned head with memory, text-LoRA, image-LoRA, dual-LoRA) on a unified all-media sticker bank. (2) A pilot benchmark showing that text-side and image-side LoRA make asymmetric contributions to exact-vs-semantic retrieval. (3) A reusable, self-contained MultiSticker codebase including data manifest construction, memory retrieval, OpenCLIP wrapping, multi-frame decoding, and SLURM-friendly run scripts.

2. Related Work

Vision-language retrieval. CLIP [6] popularised dual-encoder image-text retrieval, and OpenCLIP [3] provides open-weight reproductions trained on LAION-class corpora. Our system uses ViT-B/32 with the

laion2b-s34b-b79k weights as the backbone for both the text and image branches.

Sticker / meme retrieval. Sticker reply suggestion has been studied as a multimodal retrieval and ranking problem [4, 1]. Intent-guided sticker retrieval (IGSR) [10] adds an explicit intent classification head and a group prior. We adopt this template and add long-term memory and parameter-efficient adaptation.

Memory-augmented dialogue. Long-term dialogue memory has been used to condition open-domain chatbots [9]. We use it specifically to construct a third query stream for retrieval rather than to generate text. Our memory encoder is multilingual E5 [8], chosen for code-switched chat data.

Parameter-efficient fine-tuning. LoRA [2] adds low-rank update matrices to selected linear layers, leaving the original weights frozen. We inject LoRA into OpenCLIP’s `out_proj`, `c_fc` and `c_proj` modules through PEFT [5], and isolate text-side, image-side and dual adaptation through name-based parameter masking.

3. Method

3.1. Pipeline overview

For each turn t where the user sent a sticker s_t^* , we form three text inputs: the immediate *context* c_t (up to 12 prior turns), the retrieved *memory* m_t (concatenation of the top-3 prior session summaries), and the LLM *intent* i_t (a short description of what the sticker should express). These are encoded independently by the OpenCLIP text encoder f_T . Sticker images s are encoded by the OpenCLIP image encoder f_I (with mean-pooling over frames for animated formats). The trainable head g_θ projects the concatenation into the image space:

$$\mathbf{q}_t = \text{norm}\left(g_\theta([f_T(c_t); f_T(m_t); f_T(i_t)])\right). \quad (1)$$

Retrieval logits are $\mathbf{z}_t = \mathbf{q}_t \mathbf{B}^\top / \tau$, where $\mathbf{B} \in \mathbb{R}^{N \times d}$ is the (optionally adapted) OpenCLIP image bank and τ is a learned temperature. A second head predicts the sticker’s intent group y_t , and we minimise

$$\mathcal{L} = \text{CE}(\mathbf{z}_t, s_t^*) + \lambda \text{CE}(\mathbf{z}_t^{\text{int}}, y_t), \quad \lambda=0.2. \quad (2)$$

3.2. Long-term memory retrieval

Each raw chat file is split into sessions by a 12-hour inactivity gap. For sample t , we collect up to 8 prior sessions from the same file. Their LLM-generated session summaries are encoded with the `E5 passage: prefix`; the current session text is encoded with the `query: prefix`. After L2 normalisation, the dot products rank the prior sessions and we keep the top-3. The selected memory texts are concatenated and stored as m_t , then encoded by OpenCLIP at training time. Crucially, memory enters the model

through the *text* encoder, not as a separate vector, which lets the retrieval head learn to reweight memory against the local context.

3.3. LLM annotation

We annotate every session with a ≤ 4 -sentence `session_memory_text` (used as the passage corpus for memory retrieval), and every sticker-reply sample with one `intent_label` from a 10-class taxonomy (celebration_approval, humor_teasing, encouragement_support, gratitude, affection_care, confusion_curiosity, frustration_disapproval, disappointment_sadness, surprise_reaction, neutral_acknowledgment) plus a free-form `intent_text`. The intent label feeds the auxiliary head and induces the semantic-group label space; the intent text becomes the third query stream.

3.4. Multi-frame sticker encoding

.png stickers are encoded as one-frame clips. .gif and .webm are decoded to all frames (FFmpeg for video, PIL for GIF), each frame is encoded by the OpenCLIP image encoder, and the per-frame embeddings are mean-pooled and renormalised. During image-LoRA training, the full bank is re-encoded once per epoch without gradients; for each batch we additionally re-encode only the unique *positive* stickers with gradients (animated positives capped at 4 frames) and patch them into a cloned bank, which keeps backprop tractable for a 42K-sticker bank.

3.5. Tuning modes

We compare four tuning modes against a zero-shot CLIP baseline. **direct_clip** uses no learned head: the query is just $f_T(c_t)$ scored against the frozen image bank. **head_only** (*memory*) trains the retrieval head on top of frozen OpenCLIP encoders. **text_lora** additionally injects LoRA ($r=8$, $\alpha=16$) into the text branch. **image_lora** injects LoRA into the visual branch only, while **dual_lora** adapts both branches. Only LoRA parameters and the head are trainable; all original OpenCLIP weights are frozen.

4. Experiments

4.1. Setup

We use the U-Sticker corpus stored under `u-sticker-combined.zip`, covering 8,129 sessions and 368,503 provisional sticker-reply samples across the three media types. The full setting caps training at 100,000 samples; the validation split has 18,403 samples and the decodable sticker bank has 42,319 entries. For fast iteration we also report a *pilot* configuration (5K train, 2K val, 3K-sticker bank, 2 epochs). The text encoder is multilingual E5-small for memory retrieval and OpenCLIP ViT-B/32 (laion2b-s34b-b79k) for the CLIP branch.



Figure 1. End-to-end MultiSticker pipeline. Raw U-Sticker JSON is split into 12-hour sessions; an LLM produces a per-session memory and a per-sample intent label/text; multilingual E5 retrieves the top-3 historical sessions for the current turn; OpenCLIP independently encodes the context, memory and intent strings; the `IntentGuidedRetriever` head projects their concatenation into the CLIP image space and ranks the sticker bank by cosine similarity. An auxiliary intent classifier predicts the sticker’s group and supplies a prior for the two-stage rerank.

We use AdamW with head LR 10^{-3} and LoRA LR 10^{-4} , retrieval temperature 0.07, and intent-loss weight 0.2. All experiments use a single A100-class GPU on the cluster. Code, scripts and SLURM batch files are in `MultiSticker/scripts/`.

4.2. Metrics

The result JSONs use the legacy field names `p@K`, but because each sample has a single observed positive we interpret them as **Exact Hit@K** (the gold sticker appears in the top- K). **Group Hit@K** counts a hit if any top- K sticker shares the gold sticker’s majority intent group. **Two-stage Group Hit@30** restricts the top- K list to the two most likely intent groups predicted by the auxiliary head, then re-applies the group hit criterion. For a sticker panel UI, Hit@30 is the most informative metric; Hit@1 measures strict auto-send quality.

4.3. Results

Table 1 summarises all completed runs. Three observations stand out.

Text-side LoRA helps exact retrieval most. On both pilot and full settings, `text_lora` gives the highest Exact Hit@1 and Hit@5. This is consistent with a story where the bottleneck on the text side is domain-shifted chat language

and code-switching that the frozen OpenCLIP text encoder did not see.

Image-side LoRA helps semantic-group retrieval. `image_lora` hurts Exact Hit@K (the gold sticker often gets out-ranked by visually similar siblings) but improves both raw and two-stage Group Hit@30. The visual encoder learns to bring stickers with the same intent closer together, which is the right behaviour for a candidate-panel deployment that does not need to recover the exact id.

Dual LoRA does not always win. At pilot scale `dual_lora` gives the best Exact Hit@30 but does not beat the single-side LoRAs on their respective best metrics, consistent with mild interference between the two adaptations. The full-scale dual-LoRA run was still queued at submission time and is the next item to fill in.

5. Analysis and Discussion

Why the Exact–Group gap is so large. Across every mode, Group Hit@30 is an order of magnitude higher than Exact Hit@30. Each sample has only one observed positive even though many stickers in the bank carry the same affect, so the model is penalised every time it returns a *different but equally valid* sticker. For a deployed candidate panel this is not a real failure; we therefore recommend Group Hit@30

Mode	Trainable	Bank	Train	Exact H@1	Exact H@5	Exact H@30	Group H@30	2-stage Group H@30
<i>Full all-media run (val=18,403)</i>								
head_only (memory)	head	42,319	100K	0.0064	0.0188	0.0589	0.9578	0.7122
text_lora	head+T	42,319	100K	0.0092	0.0208	0.0565	0.9501	0.6884
image_lora	head+V	42,319	100K	0.0008	0.0029	0.0111	0.9075	0.7903
<i>Pilot run (val=2,000)</i>								
head_only (memory)	head	1,873	5K	0.0160	0.0641	0.1913	0.9710	0.7121
text_lora	head+T	1,873	5K	0.0255	0.0791	0.2359	0.9690	0.7121
image_lora	head+V	1,873	5K	0.0095	0.0461	0.1758	0.9790	0.7501
dual_lora	head+T+V	1,873	5K	0.0220	0.0736	0.2389	0.9720	0.6970

Table 1. Sticker-retrieval results. *Trainable* columns: T = text-side LoRA, V = image-side LoRA, head = IntentGuidedRetriever. The full run uses all three media types and a 42,319-sticker bank; the pilot uses 3,000-sticker cap with min frequency 2. Exact Hit@K measures recovery of the observed gold sticker; Group Hit@K measures retrieval of any sticker with the same majority intent label; 2-stage rerank restricts candidates to the top-2 predicted intent groups before applying the group criterion.

as the primary KPI and Exact Hit@1 as a strict-mode auto-reply metric.

Role of memory. Memory enters the model as a third *textual* stream rather than as a separate latent vector, which lets the head reweight memory against context per-sample. A clean ablation against a no-memory head is supported by the new `--memory-strategy disabled` flag and will be added in the next revision.

What a zero-shot CLIP baseline tells us. The `direct_clip` script ranks stickers using only the frozen OpenCLIP text encoder applied to the raw context. Any margin between it and `head_only` isolates the contribution of the head and the memory/intent streams.

Limitations. Intent labels come from a single LLM (Qwen2.5-32B-GPTQ); label noise propagates into the auxiliary loss and the group-level metrics. Memory retrieval is bounded to the same chat file, so cross-channel patterns are not exploited. Exact Hit@K is conservative; a multi-positive evaluation that accepts any same-group sticker is left for future work.

6. Conclusion

We presented MultiSticker, an intent-guided dialogue-aware sticker retrieval system that combines OpenCLIP, retrieved long-term memory, and LLM-generated intent into a single trainable head with optional LoRA adaptation on either encoder branch. On U-Sticker, text-side LoRA most improves exact recovery while image-side LoRA most improves semantic-group recovery, suggesting the two modalities should be adapted with different objectives in mind. The released code makes the five-row comparison reproducible end-to-end.

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